

FERNANDO CASTRO

Mérida, Yucatán, México | [Correo](#) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Instituto Tecnológico de Software

Mérida, Yucatán

TSU in Software Development → Bachelor of Engineering in Software Development

2025 – 2029

- Entering the dual work-study program starting September 2026.
- Relevant coursework: Software Architecture, Databases, Applied Programming, Data Structures.

EXPERIENCE

Store Manager & Internal Automation Lead

2024 – Present

AlfaMascotas — 4-branch pet-store chain

Mérida, MX

- Manage customer service and lead onboarding and training of new employees across a four-branch operation.
- Identified an operational bottleneck and built an in-house AI automation (see *Bank Transfer Verification* project) that eliminated **~2 hours/day** of manual bank-transfer reconciliation.

PROJECTS

GymTracker | *ASP.NET Core, EF Core, PostgreSQL, Docker, AWS, Terraform, xUnit*

[GitHub Demo](#)

- Built a full-stack training tracker with ASP.NET Core 10 MVC, EF Core and PostgreSQL following a layered multi-project architecture documented with ADRs.
- Engineered an **LLM application pipeline** (AI coach + context-aware chatbot) with SQL retrieval, guardrails, prompt caching and a Claude→Gemini fallback.
- **xUnit tests** and set up a GitHub Actions CI pipeline running the suite on every push and pull request.
- Deployed to **AWS** (EC2 + RDS PostgreSQL, ECR) provisioned with Terraform (IaC) and OIDC, chaining deployment to passing tests.

Bank Transfer Verification | *n8n, Claude Vision, Telegram, IMAP, Google Sheets*

[GitHub](#)

- Designed and built a 16-node n8n automation that verifies bank transfers in real time, using Claude Vision OCR checked against the bank inbox (IMAP), with a Telegram bot interface and Google Sheets logging.
- Cut manual reconciliation from **~2 h/day to near-zero** across a 4-branch operation, with responses under 30 seconds; presented to and approved by the stakeholder.

Market Backend Demo | *Java 21, Spring Boot, Spring Data JPA, MapStruct, PostgreSQL, Swagger*

[GitHub](#)

- Built a REST API for a market domain (products, categories, clients, purchases) with a layered domain/persistence (ports & adapters) architecture.
- Used Spring Data JPA and MapStruct for domain↔persistence mapping, with composite keys and cascade persistence; documented and exercised through OpenAPI/Swagger UI.

TECHNICAL SKILLS

Languages: Java, C#, Python, SQL

Backend & Frameworks: REST APIs, ASP.NET Core, Spring Boot, Entity Framework Core, Spring Data JPA.

Cloud & DevOps: AWS (EC2, RDS, ECR), Terraform (IaC), Docker, GitHub Actions (CI/CD), Git

AI & Automation: LLM application pipelines, n8n, Claude Vision, MCP, prompt & context engineering.

Databases & Docs: PostgreSQL, xUnit, OpenAPI / Swagger

CERTIFICATIONS & ADDITIONAL DETAILS

AWS: AWS Academy Graduate — Cloud Foundations & Cloud Operations

Cisco: Python Essentials 1 · Network Technician Career Path

DataCamp: Intermediate SQL · Introduction to SQL · Introduction to Docker · Intermediate Git

Languages: English · Spanish (Native)

Honors & Community: Global Game Jam Participant · Invent For The Planet Participant · IEEE Student Member